

SAND POINT 1 ENE, AK, USA

WMO: 703150

Lat: 55.3473N

Lon: 160.4660W

Elev: 240

StdP: 14.57

Time Zone: -9.00 (NAK)

Period: 09-19

WBAN: 25630

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.0	14.8	3.6	6.7	12.3	7.2	8.0	16.3	28.7	34.4	23.2	29.3	13.3	N/A	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	62.3	56.5	59.7	55.0	57.7	53.8	57.4	60.7	56.0	58.7	54.9	57.2	4.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.9	67.1	58.2	54.6	64.1	56.9	53.4	61.2	55.7	24.8	61.2	23.8	58.6	23.1	57.4	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.9	21.5	18.3	DB	11.2	67.4	4.7	4.3	7.8	70.5	5.0	73.0	2.3	75.4	-1.1	78.5
			WB	10.2	60.4	4.5	2.8	6.9	62.4	4.3	64.1	1.8	65.6	-1.5	67.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	32.6	32.6	31.1	35.7	41.3	47.4	51.7	53.3	49.2	43.1	37.2	33.7
	DBStd	9.29	6.65	6.56	7.18	5.00	3.91	3.83	3.30	3.15	3.63	4.38	5.58	6.43
	HDD50	3597	540	488	586	429	272	97	18	6	56	214	385	506
	HDD65	8841	1005	908	1051	879	736	529	412	363	474	678	835	971
	CDD50	233	0	0	0	0	1	18	72	109	32	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	7.6	8.9	9.0	7.6	8.4	7.5	6.7	4.9	5.5	6.6	8.0	8.7	9.4
	PrecAvg	41.4	3.9	3.5	2.8	2.3	2.6	3.0	3.7	3.0	4.4	3.8	4.0	4.4
(15) Precipitation	PrecMax	56.3	6.3	7.3	5.8	3.7	4.9	4.9	5.7	4.4	7.2	6.5	6.6	9.4
	PrecMin	31.1	1.6	1.1	0.8	0.7	0.7	1.3	1.5	1.1	3.0	1.9	1.5	1.6
	PrecStd	6.1	1.3	1.5	1.3	0.8	1.0	0.9	1.1	0.8	1.1	1.2	1.3	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.0	42.5	43.7	47.9	55.0	64.9	65.1	65.8	60.3	53.4	49.0	45.4	(19)
	0.4% MCWB	40.9	40.7	39.6	42.4	46.4	55.8	57.8	58.8	54.4	49.6	47.0	44.9	(20)
	2% DB	41.5	41.4	41.5	44.5	51.1	59.0	61.5	62.7	57.4	51.6	47.6	43.4	(21)
	2% MCWB	40.1	40.1	39.3	40.2	44.6	52.0	56.0	57.5	53.9	48.9	46.5	42.7	(22)
	5% DB	40.4	40.6	40.5	42.5	48.3	56.0	59.0	60.2	55.8	50.4	46.2	42.1	(23)
	5% MCWB	39.0	39.3	38.3	39.5	43.6	50.4	54.7	55.8	52.9	48.0	44.8	41.1	(24)
	10% DB	39.5	39.7	39.1	41.3	46.4	53.4	56.8	58.1	54.4	49.1	44.3	40.9	(25)
	10% MCWB	37.9	38.3	36.9	39.0	42.9	49.2	53.3	54.6	51.8	46.8	42.7	39.5	(26)
	0.4% WB	42.3	41.9	41.4	43.3	47.9	56.6	59.1	60.2	57.2	51.4	48.3	45.5	(27)
	0.4% MCDB	42.5	42.1	41.8	46.4	52.1	64.5	63.0	64.3	57.9	51.9	49.2	46.0	(28)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	40.8	40.9	40.4	41.4	46.0	53.3	56.7	58.3	55.5	50.0	47.0	43.1	(29)
	2% MCDB	41.2	41.3	41.2	43.0	49.6	58.1	60.6	61.7	56.3	51.0	47.4	43.2	(30)
	5% WB	39.4	40.0	38.9	40.5	44.6	51.2	55.3	56.5	53.8	48.6	45.2	41.4	(31)
	5% MCDB	39.9	40.4	40.2	42.1	47.5	55.2	58.5	59.1	54.9	49.9	46.1	41.7	(32)
	10% WB	38.3	38.8	37.3	39.4	43.6	49.9	54.1	55.3	52.4	47.2	43.5	39.5	(33)
	10% MCDB	39.2	39.6	38.8	41.0	45.9	53.3	56.5	57.5	54.0	48.7	44.6	40.4	(34)
	MDBR	5.9	6.3	7.6	7.4	8.0	8.8	8.0	8.3	7.8	6.7	6.5	6.0	(35)
	5% DB MCDBR	5.6	5.8	7.8	8.7	11.4	14.4	11.3	11.2	9.2	7.0	6.7	6.2	(36)
	5% DB MCWBR	5.5	5.8	6.2	6.5	7.1	9.0	7.6	7.3	6.4	6.0	6.6	6.7	(37)
	5% WB MCDBR	5.4	5.5	6.8	8.1	9.1	12.8	10.4	9.7	7.4	6.3	6.7	6.5	(38)
	5% WB MCWBR	5.5	6.0	6.1	6.6	6.4	8.3	7.3	6.8	6.3	6.0	7.1	7.2	(39)
(40) Clear-Sky Solar Irradiance	taub	0.278	0.326	0.349	0.380	0.385	0.386	0.370	0.379	0.362	0.342	0.323	0.283	(40)
	taud	2.400	2.312	2.260	2.237	2.293	2.370	2.482	2.437	2.417	2.330	2.184	2.339	(41)
	Ebn at Noon	214	236	258	266	272	276	276	265	253	226	177	179	(42)
	Edh at Noon	16	24	32	38	38	36	32	31	28	24	19	13	(43)
(44) All-Sky Solar Radiation	RadAvg	161	329	753	1147	1411	1517	1354	1124	811	460	206	112	(44)
	RadStd	21	38	117	69	170	174	145	105	70	35	20	12	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	+2.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page